D-21-05

FED SUP CLASS 5305

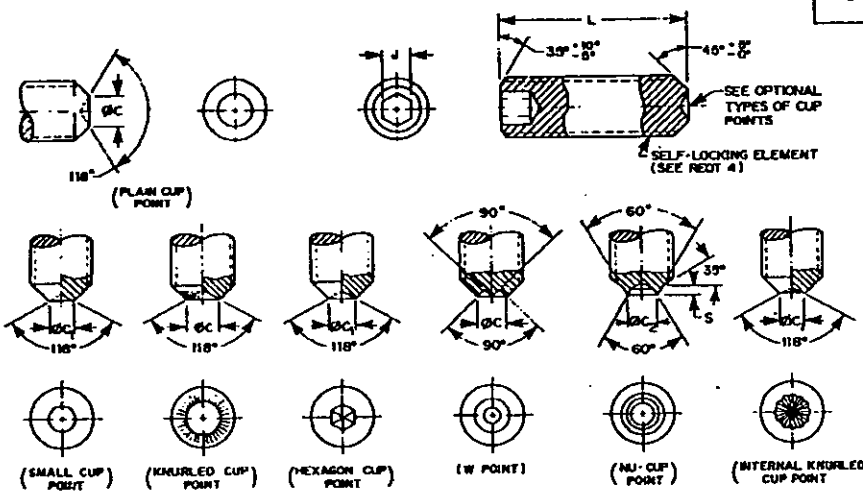


TABLE I. DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE			.060	.086	.112	.138	.164	.190
THREADS PER INCH			80 UNF-3A	64 UNF-3A	48 UNF-3A	40 UNF-3A	36 UNF-3A	32 UNF-3A
J	SOCKET WIDTH ACROSS FLATS	MIN	.028	.035	.050	.062	.078	.094
C	POINT	MAX	.033	.047	.061	.074	.087	.102
		MIN	.027	.039	.051	.064	.076	.088
C_1	POINT	MAX	.032	.043	.056	.069	.082	.095
		MIN	.027	.038	.051	.062	.074	.086
C_2	POINT	MAX	.027	.043	.059	.074	.090	.101
		MIN	.022	.038	.054	.068	.084	.095
S	POINT LENGTH	MAX	.007	.010	.013	.017	.021	.024
		MIN	.004	.007	.008	.012	.016	.019

L LENGTH	TOLERANCE	DASH NUMBER						
		PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	PLAIN	SELF-LOCKING
.125	±.010	1	9 1/	17	27	37		120 1/
.188		2	10 1/	18	28	38	48	121
.250		3	11 1/	19	29	39	49	122
.312				20	30	40	50	123
.375				21	31	41	51	124
.438						42 1/	52 1/	125 1/
.500						43	53	126
.562								127 1/
.625							54	128
.750	±.020						55	129
1.000							56	130
1.250							57	

1/ INACTIVE FOR NEW DESIGN AFTER 29 DECEMBER 1972

Ⓔ ENTIRE STANDARD REVISED

APPROVED 29 JAN 1964 REVISED Ⓔ 15 SEPT 88

USER ACTIVITIES: NAVY - MC, OS, SH

REVIEWER ACTIVITIES: ARMY - AT, DLA - IS, NSA - NS

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Neiveion for all new engineering and design applications and for reprotive use shall be made from this document when applicable.

AMSC N/A

PA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
ARMY - AR Other Cust AIR FORCE - 99		SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, CADMIUM PLATED, UNF-3A, PLAIN AND SELF-LOCKING	MS51964
PROCUREMENT SPECIFICATION FF-S-200	SUPERSEDES: MS51018, MS51020, BCTX4 AND IN PART MS18063, BCTX1, 1.1, 1.2, 3 AND 5, BCU35 AND 5.1		PAGE 1 OF 3

DD FORM 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5305-1716

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS 5305

TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

	NOMINAL SIZE		.250	.3125	.375	.4375	.500
	THREADS PER INCH		28 UNF-3A	24 UNF-3A	24 UNF-3A	20 UNF-3A	20 UNF-3A
J	SOCKET WIDTH ACROSS FLATS	MIN	.125	.156	.188	.219	.250
ØC	POINT	MAX MIN	.172 .118	.172 .156	.212 .194	.252 .232	.291 .270
$ØC_1$	POINT	MAX MIN	.125 .114	.156 .144	.187 .174	.218 .204	.250 .235
$ØC_2$	POINT	MAX MIN	.156 .158	.190 .185	.241 .236	.286 .282	.333 .328
S	POINT LENGTH	MAX MIN	.027 .022	.038 .033	.041 .036	.047 .042	.054 .049

L LENGTH	TOLERANCE	DASH NUMBER								
		PLAIN	SELF-LOCKING	PLAIN	SELF-LOCKING	PLAIN	SELF-LOCKING	PLAIN	PLAIN	SELF-LOCKING
.188	±.010	63	131							
.250		64	132	78	144	92				
.312		65	133	79	145	93				
.375		66	134	80	146	94	156	118 1/		166
.438		67 1/	135 1/	81 1/		95 1/		119 1/		
.500		68	136	82	147	96	157	120 1/	107	167
.562			137 1/							
.625		69	138	83	148	97	158		108	168
.750	±.020	70	139	84	149	98	159		109	169
.875			140		150		160			170
1.000		71	141	85	151	99	161		110 1/	171
1.250		72	142	86	152	100	162		111 1/	172
1.500			143		153	101	163		112 1/	173
1.750					154		164			174
2.000					155		165			175
2.500	±.031									176
3.000										177

1/ DIRECTIVE FOR NEW DESIGN AFTER 29 DECEMBER 1972

TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

	NOMINAL SIZE		.625	.750	.875	1.000 2/
	THREADS PER INCH		18 UNF-3A	16 UNF-3A	14 UNF-3A	14 UNS-3A
J	SOCKET WIDTH ACROSS FLATS	MIN	.312	.375	.500	.562
ØC	POINT	MAX MIN	.371 .347	.450 .425	.530 .502	.609 .579
$ØC_1$	POINT	MAX MIN	.312 .295	.375 .357	.437 .418	.500 .486
$ØC_2$	POINT	MAX MIN	.425 .428	.523 .518	——	——
S	POINT LENGTH	MAX MIN	.067 .062	.081 .076	——	——

L LENGTH	TOLERANCE	DASH NUMBER			
		PLAIN	PLAIN	PLAIN	PLAIN
.250	±.018				
.312					
.375					
.438					
.500		253	268		
.625		254	269		
.750	±.028	255	270	283	295 2/
.875		256	271	284	296 2/
1.000		257	272	285	297 2/
1.250		258	273	286	298 2/
1.500		259	274	287	299 2/
1.750		260	275	288	300 2/
2.000		261	276	289	301 2/
2.500	±.031	262	277	290	302 2/
3.000		263	278	291	303 2/

2/ DIRECTIVE FOR NEW DESIGN AFTER 30 JUNE 1988

USER ACTIVITIES: NAVY - MC, OS, SH

REVIEWER ACTIVITIES: ARMY - AT, DLA - IS, NSA - NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Recipients for all new engineering and design applications and for reprodutive use shall be made from this document when applicable.

APPROVED 29 JAN 1964 REVISED (E) FOR CHANGES SEE PAGES 1,2 AND 3

P.A. ARMY - AR Other Cust AIR FORCE - 99	INTERNATIONAL INTEREST	TITLE: SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, CADMIUM PLATED, UNF-3A, PLAIN AND SELF-LOCKING	MILITARY STANDARD MS51964
PROCUREMENT SPECIFICATION: FF-S-200	SUPERSEDES: MS51018, MS51020, BCTX4 AND IN PART MS18063, BCTX1, 1.1, 1.2, 3 AND 5, BCUX5 AND 5.1		PAGE 2 OF 3

DD FORM 1 MAY 72 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5305-1716

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.

FED. SUP CL. ASS
5305

REQUIREMENTS:

1. MATERIAL: ALLOY STEEL IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

2. PROTECTIVE COATING AND SURFACE TREATMENT: CADMIUM PLATED IN ACCORDANCE WITH PROCUREMENT SPECIFICATON.

3. THREADS: THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS FOR SIZES .190 AND SMALLER SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 21. ACCEPTABILITY OF SCREW THREADS FOR SIZES .250 AND LARGER SHALL BE IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 22. SELF-LOCKING SCREWS SHALL BE INSPECTED PRIOR TO INSTALLATION OF LOCKING ELEMENT.

4. SELF-LOCKING ELEMENT:

   (a). THE SELF-LOCKING ELEMENT SHALL BE IN ACCORDANCE WITH THE PROCUREMENT DOCUMENT.

   (b). MAXIMUM RING GAGE DIAMETER THAT LOCKING REGION OF SETSCREW MUST PASS THRU FREELY OR WITH FINGER PRESSURE SHALL BE THE NOMINAL SIZE PLUS 0.010.

   (c). MIL-F-18240 DOES NOT ESTABLISH TORQUE REQUIREMENTS FOR LENGTHS ABOVE THE DASHED LINE. ONLY A POSITIVE INDICATION OF TORQUE IS REQUIRED.

5. TOLERANCES: ANGLES ±5°, UNLESS OTHERWISE SPECIFIED.

6. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE I.

   EXAMPLE: MS51964-1
   - DASH NUMBER
   - BASIC MS NUMBER

   MS51964-1 INDICATES - SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, CADMIUM PLATED; .060-80 UNF-3A; LENGTH .125; PLAIN.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

INTERCHANGEABILITY RELATIONSHIP

SCREWS COVERED BY THE DASH NUMBERS GIVEN IN MS51018 AND MS51020 ARE CANCELLED AFTER 29 JANUARY 1964 AND SUPERSEDED BY THE SCREWS COVERED ON MS51964 HAVING THE SAME DASH NUMBERS. CANCELLED SCREWS MAY BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY MS51964 SCREWS FOR NEW DESIGN.

SCREWS COVERED BY THE DASH NUMBERS GIVEN IN MS18063 IN PART ARE CANCELLED. CANCELLED SCREWS MAY BE USED UNTIL EXISTING STOCKS ARE DEPLETED. USE ONLY THE SUPERSEDING SCREWS FOR NEW DESIGN. REPLACEMENT SHALL BE IN ACCORDANCE WITH TABLE II.

TABLE II. INTERCHANGEABILITY

CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING	CANCELLED	SUPERSEDING
MS18063	MS51964	MS18063	MS51964	MS18063	MS51964	MS18063	MS51964
DASH NUMBER		DASH NUMBER		DASH NUMBER		DASH NUMBER	
4	120	32	135	64	150	82	164
8	121	37	136	70	151	85	165
13	122	45	137	74	152	92	166
19	123	50	138	78	153	40	167
25	124	57	139	81	154	53	168
31	125	63	140	84	155	60	169
36	126	69	141	91	156	66	170
44	127	73	142	39	157	72	171
45	128	77	143	52	158	76	172
56	129	15	144	59	159	80	173
68	130	21	145	65	160	83	174
9	131	27	146	71	161	86	175
14	132	38	147	75	162	87	176
20	133	51	148	79	163	88	177
26	134	58	149				

USER ACTIVITIES: NAVY - MC, OS, SH

REVIEWER ACTIVITIES: ARMY - AT, DLA - IS, NSA - NS

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document, when applicable.

APPROVED 29 JAN 1964 REVISED (E) FOR CHANGES SEE PAGES 1, 2 AND 3

P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
ARMY - AR Other Cust AIR FORCE - 99		SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, CADMIUM PLATED, UNF-3A, PLAIN AND SELF - LOCKING	MS51964
PROCUREMENT SPECIFICATION FF - S - 200	SUPERSEDES: MS51018, MS51020, BCTX4 AND IN PART MS18063, BCTX1, 1.1, 1.2, 3 AND 5, BCUX5 AND 5.1		PAGE 3 OF 3

DD FORM 1 NOV 65 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5305-1716

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T18:14Z
Check the source to verify that this is the current version before use.